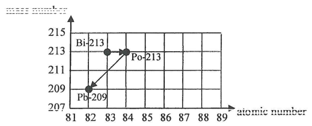
ch25 Radiation and Radioactivity

2026 Q12

12. (a) Bismuth-213 (Bi-213) is a radionuclide currently undergoing clinical trials for cancer treatment. It is prepared in liquid and injected into the human body, where it targets and destroys cancer cells. Figure 12.1 shows a decay series of Bi-213.

Figure 12.1



(i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

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*(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11}. Find its activity (in Bq) at $t = 0$ s. (2 marks)

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*(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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(b) In hospital, technicians wear protective gloves to handle liquid radioactive materials, such as those containing Bi-213. To monitor radiation exposure, they wear a specialised ring on their finger, as shown in Figure 12.2(a). Each ring contains a small chip in a plastic enclosure (Figure 12.2(b)), which detects and records radiation doses. The rings are collected every three months to monitor the radiation exposure of the technicians.

Figure 12.2(a)

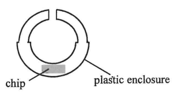

Figure 12.2(b)

(i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand ? (1 mark)

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(ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

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(iii) Which type(s) of radiation (α, β or γ) can reach the chip of the ring ? (1 mark)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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